SWIN-TOD: Smooth Wasserstein Distance and Instance-Level Neighboring Enhancement for Remote Sensing Tiny Object Detection

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***Abstract*— The advancement of deep neural network has propelled the widespread application of remote sensing target detection. However, compared to natural scenes, remote sensing targets possess inherent characteristics such as weak features and small scale, leading to a significant performance gap in traditional detection methods. To address these challenges, we undertake a systematic analysis of existing approaches, focusing on two key aspects: inadequate extraction of discriminative features and inappropriate regression measurement metrics. To tackle the first issue, an instance-level neighboring enhancement network (INEN) is proposed, enhancing the network's feature extraction capability through inter-object feature aggregation. To address the second issue, a novel metric, smooth Wasserstein loss (SWL), is devised. Building upon these principles, a new tiny object detection (TOD) network for remote sensing images is developed. Extensive experiments on AI-TOD v1/v2 and DOTA v2 remote sensing tiny target detection datasets demonstrate that our approach achieves state-of-the-art (SOTA) performance. Codes are available at https://github.com/sevenwgb/SWIN-TOD.**

***Index Terms*— Feature enhancement, graph convolutional neural network (CNN), remote sensing object detection, tiny object detection (TOD), Wasserstein distance.**

I. Introduction

REMOTE sensing object detection is a crucial research topic in industry and academia due to its widespread applications [1], [2], [3], [4]. Because of the significant variation in imaging heights and the inherent differences in target scales, objects in remote sensing images exhibit a wide range of sizes. We adhere to the definitions provided by AI-TOD [5] and COCO [6], categorizing objects based on their absolute size, defined as the square root of the area, as shown in Table I. Equipment resolution limitations and considerable distances from targets often result in numerous tiny objects with only tens or even dozens of pixels, such as airplanes, vehicles, and bridges. Detecting these tiny targets is undoubtedly crucial in many scenarios, such as large-scale monitoring, intelligent transportation, and location-based services [7], [8], [9], [10], [11].

TABLE I
SCALE LEVELS AND CORRESPONDING ABSOLUTE SIZES

Scale level	very tiny	tiny	small	medium	large
Absolute size	2–8	8–16	16–32	32–64	>64

TABLE II
DIFFERENT SCALE OBJECT RECOGNITION PERFORMANCE OF DETECTORS [12] ON COCO [6] AND AI-TOD V1 [5] DATASETS

COCO			AI-TOD v1			
AP_s	AP_m	AP_l	AP_{vt}	AP_t	AP_s	AP_m
37.7	58.4	68.1	0	10.8	28.3	38.0

Tiny objects with limited features present significant challenges for object detection tasks [13], [14], [15]. Traditional object detection methods typically focus on normal-sized objects, but experience significant degradation when tasked with detecting small-sized objects. As shown in Table II, detecting objects with recursive feature pyramid and switchable atrous convolution (DetectoRS) [12] achieves 58.4 and 68.1 average precision on medium (AP_m) and large (AP_l) objects, respectively, but only attains 37.7 average precision on small (AP_s) objects in the COCO dataset [6]. Even more, on the AI-TOD v1 dataset [5], DetectoRS only achieves 10.8 and zero average precision on tiny (AP_t) and very tiny (AP_{vt}) objects, respectively.

To further intuitively analyze the shortcomings of traditional networks in tiny object detection (TOD), we compile typical recognition results. As shown in Fig. 1, we select a representative visualization result of DetectoRS on AI-TOD v1 and DOTA v2 (with a confidence threshold of 0.5). As shown in the top-right corner of Fig. 1, we categorize the positional relationships between ground truth (GT) boxes and predicted boxes into four scenarios: 1) the intersection over union (IoU) [16] between the predicted box and the GT box is greater than 0.5 [see Fig. 1(a)]; 2) predicted boxes

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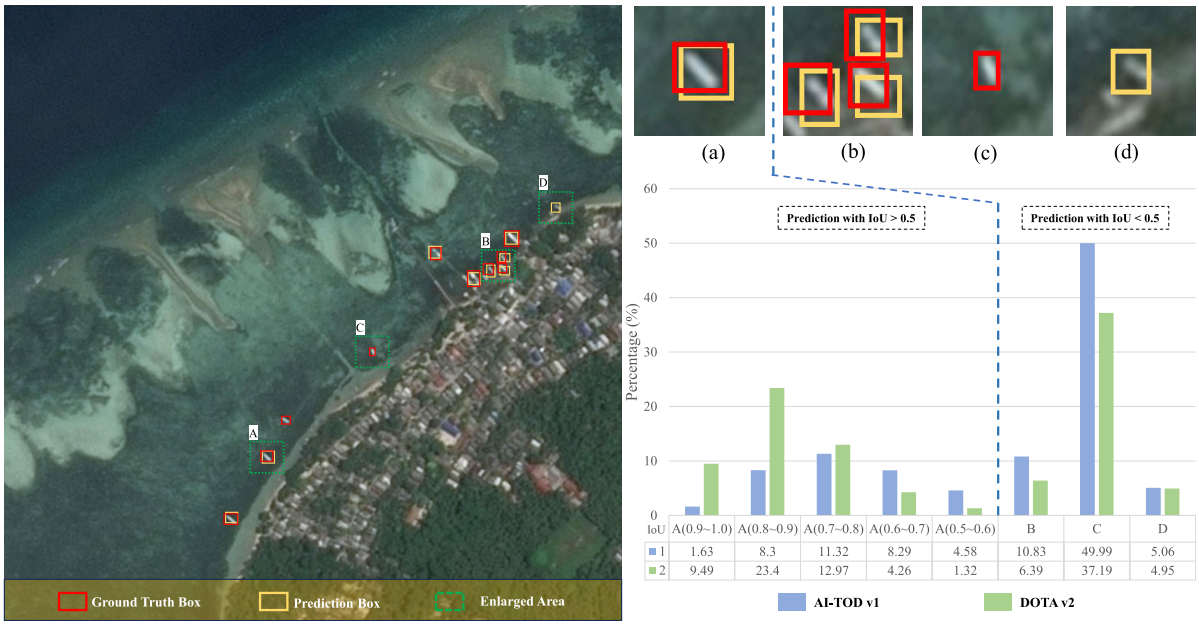


Fig. 1. Analysis of typical detection results on AI-TOD v1 and DOTA v2 datasets. The illustrative image is from AI-TOD v1. The red boxes represent GT and yellow boxes represent predicted results. (a) Detections with IoU $>$ 0.5. (b) Imprecise location detections with IoU between 0 and 0.5. (c) Missed detections with IoU $=$ 0. (d) False positives. Missed detections as in (c) are the primary reason for the poor performance, followed by imprecise location detections as in (b).

with an IoU between 0 and 0.5, which are detected but cannot be precisely located [see Fig. 1(b)]; 3) GT boxes that are completely missed, indicating that these objects are not detected by the network at all [see Fig. 1(c)]; and 4) there is no GT surrounding the predicted box [see Fig. 1(d)]. The bottom right corner shows the statistics of these scenarios. From the statistical results, we identify two main deficiencies.

1) Missed detections [see Fig. 1(c)] are the primary reason for the poor performance of the object detector. We attribute this to the inability of networks to effectively extract features from tiny objects. Specifically, tiny objects have small scale and weak features, and existing feature extraction networks based on convolutional neural network (CNN) down-sampling paradigms tend to filter out features of tiny-scale objects, making it difficult to extract discriminative features.
2) Traditional object detectors struggle with the localization regression of tiny objects, resulting in a significant number of low IoU recognition results [see Fig. 1(b)]. Locating tiny objects is more challenging than locating large objects because small perturbations of the bounding box may significantly affect the evaluation metrics of position regression (such as IoU). Additionally, false positives [see Fig. 1(d)] are relatively rare, and this study does not delve into this problem.

To address the first issue, research primarily focuses on enhancing the network's ability to extract discriminative features. In our observations, we frequently encounter a phenomenon where some objects in an image are detected while others are not. Inspired by the capability of graph convolutional networks (GCNs) to enable "continuous exchange of information/messages among nodes in a graph" [17], [18], we establish a instance-level neighboring enhancement network (INEN). We aim to facilitate information exchange among the same category instances, allowing nodes with stronger features to guide the feature enhancement of nodes with weaker features, thereby reducing the rate of missed detections.

Regarding the second issue, studies primarily focus on improving regression metrics. We propose a novel distance metric called smooth Wasserstein loss (SWL), which is inspired by smoothL1loss [19] and Wasserstein distance [20], incorporating different Wasserstein distances at various scales (first-order Wasserstein distance for large scale and second-order Wasserstein distance for small scale).

Previous GCN-based methods [21], [22] always design class-aware GCN modules. Compared to them, our method is class-agnostic and instance-aware, demonstrating better adaptability to class variations. Moreover, since tiny objects inherently have few features, our instance-aware method minimizes feature loss, which is crucial for effective detection of tiny objects. In contrast to previous distance metrics [19], [20], [23], our approach integrates the advantages of target-specific 2-D Gaussian modeling and scale-aware segmented loss calculation, accurately capturing the true distribution of targets while effectively mitigating sensitivity to outliers.

The contributions of this article are threefold.

1) We propose an INEN module, which can be easily integrated into the network to enhance the network's feature extraction capability and reduce missed detections.
2) Addressing the shortcomings of existing distance metrics, we introduce a novel metric parameter that improves the network's regression ability, alleviating the problem of inaccurate regression in detecting tiny objects.
3) We incorporate our proposed module into several networks and conduct extensive comparative experiments and ablation studies. The experimental results demonstrate the effectiveness of our method and provide benchmarks for the application of the above modules.

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II. Related Works

A. Remote Sensing Tiny Object Detection

In TOD tasks, frameworks typically extend general object detection models, emphasizing discriminative feature extraction, which is achieved through three primary methods: data augmentation, feature augmentation, and train strategy.

1) Data Augmentation: In deep neural networks, sufficient data is the cornerstone of performance. However, tiny objects typically rarely occupy in datasets, which undermines the TOD performance. Addressing this issue, researches [24], [25] propose improvements by increasing the number of target instances. Additionally, due to the small scale and weak discriminative features of tiny objects, super-resolution methods [26], [27], [28], [29] are utilized to enhance the detailed information.

2) Feature Augmentation: Effectively integrating high-level semantic information and low-level geometric details in CNNs can enhance the representation of objects. Series of methods [30], [31], [32], [33], [34] augment target features by fusing features from different layers. Additionally, targets and backgrounds exist under certain constraints. Studies [35], [36], [37] leverage this relationship to enhance target feature representation. Furthermore, researches [38], [39] design various attention modules to increase the network's attention to discriminative features.

3) Train Strategy: Detectors often struggle to achieve high performance simultaneously for both large and small objects. Correspondingly, scale normalization for image pyramids (SNIP) [40] selectively trains on instances with desired resolutions scale range, while ignoring the rest. Sniper [41] recommends sampling chips from a multiscale image pyramid for efficient training. Najibi et al. [42] proposes a coarse-to-fine pipeline for detecting small objects. Scale aware network (SAN) [43] maps features from different spaces into a scale-invariant subspace, making detectors more robust to scale variation.

In remote sensing scenes, rotational region convolutional neural network (R2CNN) [44] proposes a unified self-enhancement network to elegantly identify tiny targets in large-scale remote sensing images. Feature and spatial aligned network (FSANet) [45] proposes a feature space alignment network to obtain more discriminative features and accurate localization. Mask augmented attention feature pyramid network (MA2-FPN) [46] aggregates contextual and spatial feature information for tiny objects using a large-kernel separable convolution attention mechanism. However, existing methods rarely leverage the feature correlations between instances to enhance the extraction of discriminative features. Feature enhancement, fusion and context aware YOLO (FFCA-YOLO) [47] enhances local area awareness, multiscale feature fusion, and global association across channels and spatial dimensions with its feature enhancement, feature fusion, and spatial context aware modules. Dot distance (DotD) [48] introduces a new metric, defined by the normalized Euclidean distance between the centers of two bounding boxes. Normalized wasserstein distance and ranking-based alignment (NWD-RKA) [49] introduces normalized Wasserstein distance and a new ranking-based assigning strategy. Multi-scale feature combined with H-network (MSHNet) [14] proposes a novel scale and location sensitive loss for infrared small target detection, which helps detectors distinguish objects with different scales and locations.

B. Graph Convolutional Neural Network

Graph neural networks [50] break the traditional constraint of converting graph data into a set of vectors, enabling direct data processing on the raw graph data, which are commonly used for processing unstructured, non-Euclidean data [51], [52], [53]. GCN [54] introduces the application of CNNs to graph-structured data, generalizing the convolution operator to irregular domains, which is typically expressed as a neighborhood aggregation or message-passing scheme. Object detection with graph convolutional network (OD-GCN) [21] introduces the incorporation of inter-class relationships into object category discrimination, with relationships between objects calculated based on the probability of object occurrences in the COCO dataset. Self-attention graph convolutional network (SAT-GCN) [55] enhances semantic representation by aggregating neighborhood information and focusing on important relationships using both GCN and self-attention. Spatial-attention graph convolution network (S-ATGCN) [56] addresses the partition effect in point cloud processing by introducing a feature enhancement layer. GraphFPN [57] introduces a graph feature pyramid network that adapts its topological structure to different intrinsic image structures and supports feature interaction across all scales simultaneously. GCN achieves information exchange between different nodes through aggregation, using weighted aggregation to perform various prediction tasks [17]. Based on the aforementioned characteristics of GCN, this study proposes the INEN. INEN focuses on inter-node information propagation and facilitates the guidance and augmentation of nodes with weaker features by those possessing stronger features, consequently diminishing the occurrence of missed detections.

C. Distance Metrics in Detection Tasks

Object detection consists of two subtasks: classification and localization, corresponding to classification loss and regression loss, respectively. The former is typically measured using cross-entropy loss and its variants, such as focal loss [58] and varifocal loss [59]. The latter is used to measure the accuracy of predicted box localization and can be categorized as follows: norm-based, IoU-based, and distribution-based.

1) Norm-Based: $L_1 = |x|$ and $L_2 = |x|^2$ are used to measure regression loss based on the coordinates of bounding box corners. However, L_1 suffers from a problem of no differentiability at the origin (small scale), while L_2 is not robust enough in handling outliers (large scale). Therefore, Fast R-CNN [19] introduced smoothL1loss, which effectively leverages the advantages of L_1 and L_2 at different scales.

2) IoU-Based: Unlike norm-based metrics, IoU-based methods consider the relationships between the corner points of the box and regress them as a whole entity. Yu et al. [60] proposes IoU metric, generalized intersection over union (GIOU) [61] introduces the concept of the minimum enclosing

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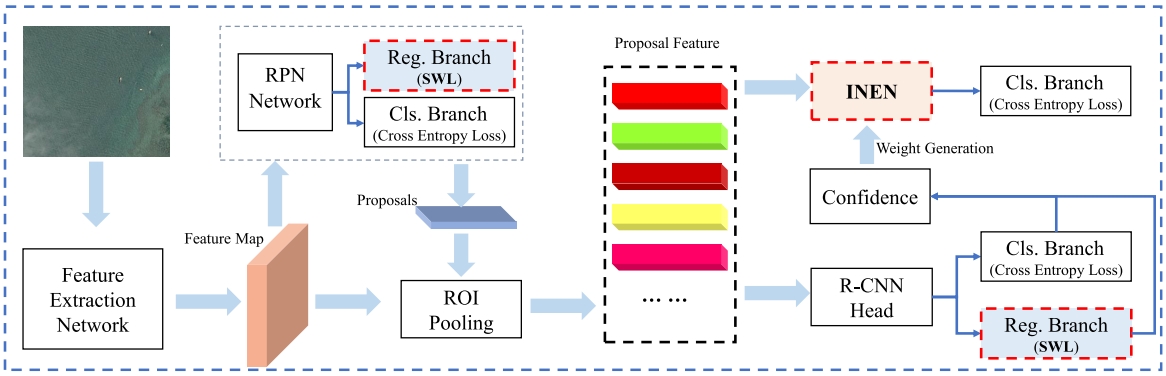


Fig. 2. Pipeline of SWIN with Faster R-CNN framework. Compared to Faster R-CNN network, the structural modifications include replacing the box regression loss (both in the RPN and the Head) with the SWL and integrating the INEN into the Head.

box, and distance intersection over union (DIOU) [62] addresses the regression convergence issue of predicting box inside the GT box. Paper [63] further improves convergence speed and addresses sample imbalance issues.

3) Distribution-Based: Background frequently appears inside bounding boxes due to the non-rectangular nature of most real objects [20], [23], [64]. NWD [20] models the target as a 2-D Gaussian distribution, where positions closer to the center contribute more, while positions farther from the center contribute less. Therefore, the NWD metric based on Wasserstein distance is proposed and improves detection performance significantly. Similarly, Gaussian receptive field based label assignment (RFLA) [23] also models the target as a 2-D Gaussian distribution and utilizes the Kullback–Leibler (KL) divergence of this distribution to develop a new label assignment method. Inspired by smoothL1loss and NWD, we introduce a new metric named SWL. Experimental results show the effectiveness of SWL.

III. Methodology

A. Network Architecture

The main contribution of this article is the introduction of two modules: the INEN and the SWL. These modules are embeddable and can be easily incorporated into existing networks such as Faster R-CNN [65], DetectoRS [12], RetinaNet [58], and others. Without loss of generality, we take Faster R-CNN as an example. Fig. 2 illustrates the pipeline of INEN and SWL within the Faster R-CNN framework. Compared to the base Faster R-CNN network, the structural modifications include replacing the box position regression loss (both in the region proposal network (RPN) and the detection Head) in the network with SWL, and integrating the INEN into the Head. Specifically, the INEN utilizes the predicted results from the base network's Head to guide the network in interacting feature information between proposals, resulting in enhanced proposal features. Subsequently, the enhanced features are used for classification. The detailed process will be elaborated in Section III-B.

As shown in Fig. 2, the network has two classification branches. We combine these two loss terms using blending coefficients α to compute the weighted sum by the following equation:

$$L_{\text{cls}} = \alpha L_{\text{cls_roihead}} + (1-\alpha) L_{\text{cls_inen}} \tag{1}$$

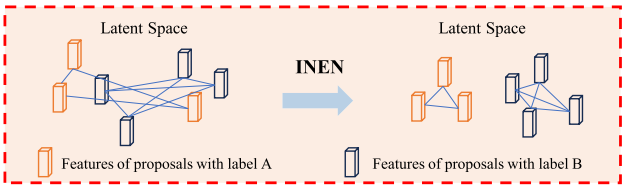


Fig. 3. Motivational schematic of the INEN. INEN enhances node features through feature aggregation.

where L_{cls}, $L_{\text{cls_roihead}}$, and $L_{\text{cls_inen}}$ represent blended total loss with the weight parameter α, classification loss of the origin Head, and classification loss of INEN, respectively.

B. Instance-Level Neighboring Enhancement Network

1) Motivation: The motivation schematic of the INEN is shown in Fig. 3. INEN enhances node features through feature aggregation. We make the Assumption 1 as the cornerstone of INEN.

Assumption 1:

1) Features of objects with the same category have similar distributions.
2) Objects with higher confidence have more informative features.
3) Objects with stronger informative features can better guide objects with weaker features for feature enhancement.
4) Features can be transferred between objects to achieve feature enhancement objectives.

In a multilayer INEN network, the inter-layer propagation is represented as follows:

$$H^{(l+1)} = \sigma\left(\widetilde{D}^{-\frac{1}{2}}\widetilde{A}\widetilde{D}^{-\frac{1}{2}}H^{(l)}W^{(l)}\right) \tag{2}$$

where $H^{(l)} \in \mathbb{R}^{N\times F}$ is the activations matrix in the lth layer, N represents the number of nodes, and F represents the feature dimension of each node. $\widetilde{A} = A+I$ is the adjacency matrix (A) of the graph with added self-connections (I). $\widetilde{D}$ is the degree matrix of $\widetilde{A}$, which can be calculated by $\widetilde{D}_{ii} = \sum_j \widetilde{A}_{ij}$. $W^{(l)}$ is the trainable weight matrix of the lth layer. $\sigma(\cdot)$ denotes an activation function, such as rectified linear unit (ReLU) or softmax.

Specifically, with $x_i^{(l+1)} \in \mathbb{R}^F$ denoting node features of node i in layer $(l+1)$ and $a_{ji} \in \mathbb{R}^{N\times N}$ denoting edge weight

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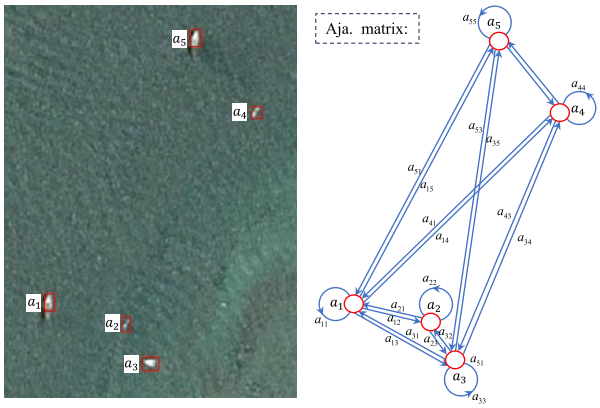


Fig. 4. Adjacency matrix diagram. The edges have direction and weight: an edge from a_i to a_j represents information transmission from a_i to a_j, with the weight of this transmission denoted as a_{ij}.

from node j to node i, message-passing graph neural networks can be described as follows:

$$x_i^{(l+1)} = \gamma^{(l)}\left(x_i^{(l)}, \oplus\phi^{(l)}\left(x_i^{(l)}, x_j^{(l)}, a_{ji}\right)\right) \tag{3}$$

where $\oplus$ denotes a differentiable, permutation invariant function, for example, sum, mean, or max; γ and ϕ denote differentiable functions such as multilayer perceptrons (MLPs). Based on the above-mentioned assumption and definition, we construct a directed weighted graph structure as Fig. 4. In the graph, a_1–a_5 are targets of the "*ship* (SH)" category. We establish edges connecting targets of the same category. The edges have direction and weight: an edge from a_i to a_j represents information transmission from a_i to a_j, with the weight of this transmission denoted as a_{ij}. In this study, a larger weight indicates stronger information transmission.

So, we represent the adjacency matrix as $A = \{a_{ij}\} \in \mathbb{R}^{N\times N}$ where N represents the number of nodes. Furthermore, we incorporate the confidence of target categories into the design of the adjacency matrix. We establish edges only between nodes with the same category, and no edges are formed between nodes of different classes (including background class, where the weight is also set to 0). For the weights of these edges, a_{ij} can be calculated by the following equations, and the confidence of nodes is used as the measurement criteria:

$$a_{ij} = \frac{2 \cdot e^{c(a_i)}}{e^{c(a_i)} + e^{c(a_j)}} \tag{4}$$

$$c(a_i) = p(a_i) \cdot \mathrm{reg}(a_i) \tag{5}$$

where $c(\cdot)$ denotes the confidence of a node, $p(\cdot)$ denotes the posterior probabilities of the node in their GT classes, and $\mathrm{reg}(\cdot)$ represents the discrepancy between the node and GT distributions, which can be calculated by distance metrics described in Section II-C. Clearly, the more similar a target label is to the GT and the smaller the distribution discrepancy, the greater the confidence. According to (4), the edge weight from this node to other nodes becomes larger, indicating more information transfer to other targets. This is consistent with the assumptions we made in constructing the INEN.

2) Proposed Module: We design the INEN for conducting instance-level feature interactions. Fig. 5 illustrates the workflow of the INEN. Specifically, the INEN has a two-layer structure, and the forward propagation formula is as follows:

$$Z = \mathrm{softmax}\left(\hat{A}\mathrm{ReLU}\left(\hat{A}XW^{(0)}\right)W^{(1)}\right) \tag{6}$$

where $\hat{A} = \widetilde{D}^{-1/2}\widetilde{A}\widetilde{D}^{-1/2}$. This network outputs posterior probability for proposal boxes across different categories. The cross-entropy loss between posterior probability and GT is calculated to train the network parameters through backpropagation. The INEN module is used to facilitate information propagation between nodes, enhancing node features. Here, we use Faster R-CNN as an example to demonstrate how this module is applied in the Head structure.

We calculate the adjacency matrix by Algorithms 1 and 2. The former is employed during training, while the latter is utilized during testing. These algorithms exhibit distinctions in the determination of proposal labels. During the training stage, GT labels are available, and we utilize the posterior probability associated with these labels to compute the adjacency matrix. Conversely, in the testing stage, where GT labels are unavailable, we employ the maximum posterior probability to calculate the adjacency matrix.

Algorithm 1 Calculating the Adjacency Matrix During Training

Input:
$p \in \mathbb{R}^{N\times C}$: the posterior probabilities of all proposals
$b \in \mathbb{R}^{N\times 4}$: the box coordinates of all proposals
$labels \in \mathbb{R}^{N}$: GT labels of all proposals
$b_{gt} \in \mathbb{R}^{N\times 4}$: the GT coordinates matched proposals
Output:
$A = \{a_{ij}\} \in \mathbb{R}^{N\times N}$: adjacency matrix.

Get the posterior probability of all proposals:
for $i = 1$ to N
$p(a_i) = p[i][labels[i]]$
end for
Calculate the $Reg(\cdot)$ for all proposals:
for $i = 1$ to N
Calculate $Reg(a_i)$ by Eq. 14
end for
Calculate $c(\cdot)$ for all proposals:
for $i = 1$ to N
Calculate $c(a_i)$ by Eq. 5
end for
Calculate a_{ij}:
Initialize $A = \{a_{ij}\}$ as zero matrix.
for $i = 1$ to N
for $j = 1$ to N
if $labels[i] == labels[j]$
Calculate a_{ij} by Eq. 4
end if
if $i == j$
$a_{ij} = 1$
end if
end for
end for

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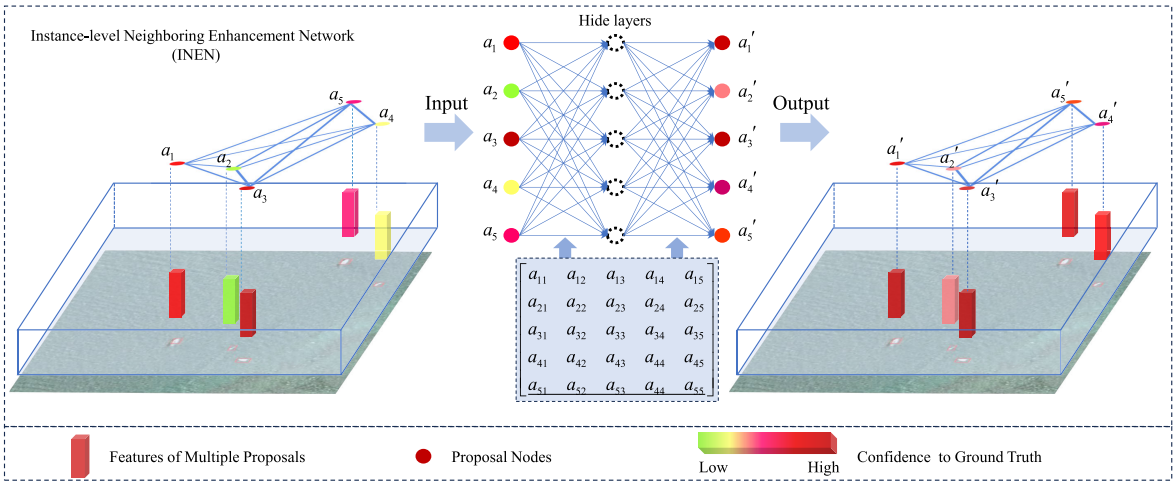


Fig. 5. Workflow of the INEN. INEN has a two-layer MLP. The prototypes of proposals are fed into the INEN, yielding aggregated features of proposals as output.

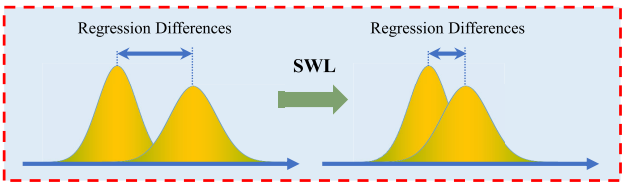


Fig. 6. Motivational schematic of the SWL. SWL enhances the network's regression capability by modeling representations of tiny objects.

During both training and testing, we first perform dimensionality reduction on the proposals' features ($\mathbb{R}^{N\times D\times H\times W}$) to obtain prototypes ($\mathbb{R}^{N\times D}$). Then, we input these prototypes and the adjacency matrix into INEN. This process yields posterior probabilities of proposals for each category. During the training phase, the network computes the cross-entropy loss between the posterior probabilities and the GT labels. Subsequently, this cross-entropy loss is used for backpropagation to optimize the network parameters. In the testing phase, the posterior probabilities are directly returned for category determination.

C. Smooth Wasserstein Loss

1) Motivation: The motivation schematic of the SWL is shown in Fig. 6. We make the following assumption as the cornerstone of SWL.

Assumption 2: In a target bounding box, the center pixel carries the highest weight, with the importance of pixels decreasing gradually from the center toward the boundaries. Therefore, the horizontal bounding box $m = (cx, cy, w, h)$ can be modeled as a 2-D Gaussian distribution $\mathcal{N}(\mu, \Sigma)$ with $\mu = [cx, cy]^{\mathrm{T}}$ and $\Sigma = \mathrm{diag}(w^2/4, h^2/4)$, where cx and cy are the center coordinates, and w and h denote the width and height, respectively.

Definition 1: The Wasserstein distance, also known as the Earth Mover's distance (EMD), measures the minimum cost required to transform distribution P into distribution Q. The kth order expectation formula for Wasserstein is as follows:

$$W(P, Q) = \left(\inf_{\gamma\in\Pi(P,Q)} \mathbb{E}_{(x,y)}\|x-y\|^k\right)^{\frac{1}{k}} \tag{7}$$

where $\Pi(P, Q)$ denotes the joint probability distribution of P and Q, $\gamma(x, y)$ represents the probability of P occurring as x, then Q occurring as y, and inf represents the lower bound. Furthermore, the similarity between bounding boxes $m = (cx, cy, w, h)$ and $m' = (cx', cy', w', h')$ can be converted to the distribution distance between two Gaussian distributions $[m \sim \mathcal{N}(\mu, \Sigma)$ and $m' \sim \mathcal{N}(\mu', \Sigma')]$. The first-order and second-order Wasserstein distance between these two boxes is defined in the following equations, respectively,

$$W_1(m, m') = |cx' - cx| + |cy' - cy| + \left|\frac{w'}{2} - \frac{w}{2}\right| + \left|\frac{h'}{2} - \frac{h}{2}\right| \tag{8}$$

$$W_2^2(m, m') = (cx' - cx)^2 + (cy' - cy)^2 + \left(\frac{w'}{2} - \frac{w}{2}\right)^2 + \left(\frac{h'}{2} - \frac{h}{2}\right)^2. \tag{9}$$

Definition 2 [Offset Vector ($\vec{OV}$)]: The distance from one point to another in a coordinate system. The $\vec{OV}$ of m and m' in $\mathbb{R}^{CX\times CY\times W\times H}$ can be calculated by the following equation:

$$\vec{OV}_R = \begin{bmatrix} cx'-cx \\ cy'-cy \\ \frac{w'}{2} - \frac{w}{2} \\ \frac{h'}{2} - \frac{h}{2} \end{bmatrix}. \tag{10}$$

Therefore, (8) and (9) can be simplified to (10) and (11)

$$W_1(m, m') = \left\|\vec{OV}_R\right\|_1 \tag{11}$$

$$W_2^2(m, m') = \left\|\vec{OV}_R\right\|_2. \tag{12}$$

The first-order Wasserstein distance is the l_1-norm (Manhattan norm) of $\vec{OV}_R$, and the second-order Wasserstein distance is the l_2-norm (Euclidean norm) of $\vec{OV}_R$.

NWD [20] and RFLA [23] both use the second-order Wasserstein distance as the basis for measuring the distance between two boxes. The derivative of the second-order Wasserstein distance is as follows:

$$\frac{\partial W_2^2(m, m')}{\partial \vec{OV}_R} = 2 \times \vec{OV}_R. \tag{13}$$

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Algorithm 2 Calculating the Adjacency Matrix During Testing

Input:
$p \in \mathbb{R}^{N\times C}$: the posterior probabilities of all proposals

Output:
$A = \{a_{ij}\} \in \mathbb{R}^{N\times N}$: adjacency matrix.

Get the Pseudo-labels of all proposals $labels \in \mathbb{R}^N$**:**
for $i = 1$ to N
$labels[i] = argmax(p[i])$
end for

Get the posterior probability of all proposals:
for $i = 1$ to N
$p(a_i) = p[i][labels[i]]$
end for

Calculate $c(\cdot)$ **for all proposals:**
for $i = 1$ to N
$c(a_i) = p(a_i)$
end for

Calculate a_{ij}**:**
Initialize $A = \{a_{ij}\}$ as zero matrix.
for $i = 1$ to N
for $j = 1$ to N
if $labels[i] == labels[j]$
Calculate a_{ij} by Eq. 4
end if
if $i == j$
$a_{ij} = 1$
end if
end for
end for

Similar to $L2$ loss, the second-order Wasserstein distance is also sensitive to outliers. Additionally, the first-order Wasserstein distance has a problem of no differentiability at the origin. Based on the above analysis and inspired by the smoothL1loss, we design the SWL as follows:

$$\mathrm{SWL} = \begin{cases} \left\|\vec{OV}_R\right\|_1 - 0.5 * \mathrm{Thres}, & \left\|\vec{OV}_R\right\|_1 > \mathrm{Thres} \\ 0.5 * \left\|\vec{OV}_R\right\|_2 / \mathrm{Thres}, & \text{otherwise} \end{cases} \tag{14}$$

where Thres is a hyperparameter threshold with a value greater than 0.

2) Proposed Module: For box regression loss, we utilize the previously introduced SWL, which is used similar to the common smoothL1loss. As a hyperparameter, the threshold Thres needs to be adjusted according to specific circumstances. SWL exhibits excellent transferability and adaptability, suitable for measuring the distance or difference between two boxes in various contexts, such as box loss in the RPN and the Head module.

IV. Experiments and Analysis

A. Implementation Details

1) Datasets: We evaluate the proposed method on three popular TOD benchmark datasets: AI-TOD v1 [5], AI-TOD v2 [49], and DOTA v2 [66]. AI-TOD v1 includes 28 036 remote sensing images with 700 621 objects across eight classes. AI-TOD v2 expands on AI-TOD v1 with the same classes and images but increases annotated instances to 752 745. DOTA v2 comprises 11 268 aerial images with about 1.8 million object instances spanning 18 categories. These datasets contain a large number of tiny objects, especially AI-TOD v1/v2, where the average absolute size is only 12.8/12.7 pixels, which is significantly smaller than other remote sensing datasets. As mentioned in [49], the AI-TOD v1 dataset suffers from some missed annotations, mainly due to its reliance on publicly available aerial image datasets that are not specifically designed for TOD tasks. In contrast, AI-TOD v2 has a higher annotation quality. Therefore, in subsequent ablation studies, unless otherwise specified, we default to conducting experiments on AI-TOD v2. In our experiments, the training was conducted on the "trainval" dataset for AI-TOD v1 and AI-TOD v2, the "train" dataset for DOTA v2, while all testing was performed on the "test" dataset.

2) Parameter Settings: Our experiments are conducted on four Tesla P100 GPUs with 16-GB memory. The code is developed based on the MMDetection [67]. We use the ImageNet pretrained ResNet-50 [68] with FPN [30] as the backbone. All models are trained using the stochastic gradient descent (SGD) optimizer for 12 epochs with 0.9 momenta, 0.0001 weight decay, and 4 batch sizes. We set the initial learning rate as 0.01 and decay it at epochs 8 and 11. PyTorch Geometric (PyG)[1] library is used in our INEN module. In the experiments with YOLOv5[2] and YOLOv9,[3] we adhere to the official code configurations and conduct training for 300 epochs. The aforementioned training and inference parameters are consistent across all experiments, unless explicitly stated otherwise. More details can be obtained from our code repository.

3) Evaluation Metric: We use the average precision metric to evaluate the performance of the proposed method. Specifically, $AP_{0.5}$ means the IoU threshold of true positive (TP) is 0.5, $AP_{0.75}$ means the IoU threshold of defining TP is 0.75, AP means the average value from $AP_{0.5}$ to $AP_{0.95}$, with IoU interval of 0.05. Note that $AP_{0.5}$, $AP_{0.75}$, and AP take objects of all scales into consideration. Moreover, AP_{vt}, AP_t, AP_s, and AP_m are for very tiny, tiny, small, and medium scale evaluation in Table I.

B. Main Result

1) Comparison With Previous Methods: To reveal the effectiveness of SWIN, we conduct experiments on AI-TOD v1, AI-TOD v2, and DOTA v2 datasets. As shown in Tables III–V, SWIN can be embedded into various detectors, such as one-stage and multistage detectors.

As shown in Table III, DetectoRS w/ RFLA w/ SWIN achieves 25.9@AP and 58.3@$AP_{0.5}$ on AI-TOD v1, surpassing the state-of-the-art (SOTA) competitors by 1.1 and 3.1 points, respectively. Similarly, the application of SWIN to other models also results in AP improvements ranging from 0.6 to 3.4. As shown in Table IV, DetectoRS w/ RFLA w/ SWIN achieves 26.5@AP and 61.5@$AP_{0.5}$ on AI-TOD v2, outperforming the

[1]https://pytorch-geometric.readthedocs.io/en/latest/index.html
[2]https://github.com/ultralytics/yolov5
[3]https://github.com/WongKinYiu/yolov9

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TABLE III

RESULTS ON AI-TOD V1. NOTE THAT MODELS ARE TRAINED ON THE "TRAINVAL" SET AND VALIDATED ON THE "TEST" SET. FCOS* MEANS USING P2-P6 OF FPN. THE BEST SCORES ARE HIGHLIGHTED IN BOLD. GRAY ROWS REPRESENT THE RESULTS WITH SWIN

Method	Backbone	AP	$AP_{0.5}$	$AP_{0.75}$	AP_{vt}	AP_t	AP_s	AP_m
TridentNet [32]	ResNet-50	0.75	20.9	3.6	1.0	5.8	12.6	14.0
DotD [48]	ResNet-50	16.1	39.2	10.6	8.3	17.6	18.1	22.1
Faster R-CNN [65]	ResNet-50	11.1	26.3	7.6	0.0	7.2	23.3	33.6
Faster R-CNN w/ SWIN	ResNet-50	$14.0^{(\uparrow 2.9)}$	$30.0^{(\uparrow 3.7)}$	$10.8^{(\uparrow 3.2)}$	$4.2^{(\uparrow 4.2)}$	$8.6^{(\uparrow 1.4)}$	$21.2^{(\downarrow 2.1)}$	$34.3^{(\uparrow 0.7)}$
Faster R-CNN w/RFLA [23]	ResNet-50	21.1	51.6	13.1	9.5	21.2	26.1	31.5
Faster R-CNN w/RFLA w/ SWIN	ResNet-50	$23.3^{(\uparrow 3.2)}$	$55.5^{(\uparrow 3.9)}$	$16.6^{(\uparrow 3.5)}$	$11.9^{(\uparrow 2.4)}$	$22.3^{(\uparrow 1.1)}$	$27.3^{(\uparrow 1.2)}$	$32.5^{(\uparrow 1.0)}$
Cascade R-CNN [69]	ResNet-50	13.8	30.8	10.5	0.0	10.6	25.5	36.6
Cascade R-CNN w/ SWIN	ResNet-50	$15.7^{(\uparrow 1.9)}$	$33.4^{(\uparrow 2.6)}$	$12.6^{(\uparrow 2.1)}$	$4.8^{(\uparrow 4.8)}$	$11.3^{(\uparrow 0.7)}$	$25.7^{(\uparrow 0.2)}$	$36.0^{(\downarrow 0.6)}$
Cascade R-CNN w/RFLA [23]	ResNet-50	22.1	51.6	15.6	8.2	22.0	27.3	35.2
Cascade R-CNN w/RFLA w/ SWIN	ResNet-50	$24.2^{(\uparrow 2.1)}$	$54.4^{(\uparrow 2.8)}$	$17.3^{(\uparrow 1.7)}$	$10.6^{(\uparrow 2.4)}$	$24.1^{(\uparrow 2.1)}$	$27.8^{(\uparrow 0.5)}$	$35.6^{(\uparrow 0.4)}$
DetectoRS [12]	ResNet-50	14.8	32.8	11.4	0.0	10.8	28.3	38.0
DetectoRS w/ SWIN	ResNet-50	$16.2^{(\uparrow 1.4)}$	$36.7^{(\uparrow 3.9)}$	$13.2^{(\uparrow 1.8)}$	$1.6^{(\uparrow 1.6)}$	$12.1^{(\uparrow 1.3)}$	$28.2^{(\downarrow 0.1)}$	$38.2^{(\uparrow 0.2)}$
DetectoRS w/RFLA [23]	ResNet-50	24.8	55.2	18.5	9.3	24.8	30.3	38.2
DetectoRS w/RFLA w/ SWIN	ResNet-50	$\mathbf{25.9}^{(\uparrow 1.1)}$	$\mathbf{58.3}^{(\uparrow 3.1)}$	$\mathbf{19.2}^{(\uparrow 0.7)}$	$\mathbf{10.8}^{(\uparrow 1.5)}$	$\mathbf{25.9}^{(\uparrow 1.1)}$	$\mathbf{30.9}^{(\uparrow 0.6)}$	$\mathbf{39.1}^{(\uparrow 0.9)}$
SSD-512 [70]	VGG-16	7.0	21.7	2.8	1.0	4.7	11.5	13.5
PAA [71]	ResNet-50	10.0	26.5	6.7	3.5	10.5	13.1	22.1
ATSS [72]	ResNet-50	12.8	30.6	8.5	1.9	11.6	19.5	29.2
RetinaNet [58]	ResNet-50	8.7	22.3	4.8	2.4	8.9	12.2	16.0
RetinaNet w/ SWIN	ResNet-50	$12.1^{(\uparrow 3.4)}$	$24.1^{(\uparrow 1.8)}$	$7.9^{(\uparrow 3.1)}$	$8.6^{(\uparrow 6.2)}$	$13.4^{(\uparrow 4.5)}$	$12.6^{(\uparrow 0.4)}$	$17.2^{(\uparrow 1.2)}$
RetinaNet w/RFLA [23]	ResNet-50	9.1	23.1	5.2	4.1	10.5	10.5	12.3
RetinaNet w/ RFLA w/ SWIN	ResNet-50	$10.6^{(\uparrow 1.5)}$	$23.8^{(\uparrow 0.7)}$	$6.3^{(\uparrow 1.1)}$	$6.2^{(\uparrow 2.1)}$	$12.1^{(\uparrow 1.6)}$	$12.6^{(\uparrow 2.1)}$	$13.1^{(\uparrow 0.8)}$
RepPoints [73]	ResNet-50	9.2	23.6	5.3	2.5	9.2	12.9	14.4
OTA [74]	ResNet-50	10.4	24.3	7.2	2.5	11.9	15.7	20.9
AutoAssign [75]	ResNet-50	12.2	32.0	6.8	3.4	13.7	16.0	19.1
AutoAssign w/RFLA [23]	ResNet-50	14.2	37.8	6.9	6.4	14.9	17.4	21.8
M-CenterNet [5]	DLA-34	14.5	40.7	6.4	6.1	15.0	19.4	20.4
FCOS [76]	ResNet-50	12.6	30.4	8.1	2.3	12.2	17.2	25.0
FCOS* [76]	ResNet-50	15.4	36.3	10.9	6.0	17.6	18.5	20.7
FCOS* w/ SWIN	ResNet-50	$16.2^{(\uparrow 0.8)}$	$38.5^{(\uparrow 2.2)}$	$11.4^{(\uparrow 0.5)}$	$8.1^{(\uparrow 2.1)}$	$19.5^{(\uparrow 1.9)}$	$19.3^{(\uparrow 0.8)}$	$21.3^{(\uparrow 0.6)}$
FCOS* w/RFLA [23]	ResNet-50	16.3	39.1	11.3	7.3	18.5	19.8	21.8
FCOS* w/RFLA w/ SWIN	ResNet-50	$16.9^{(\uparrow 0.6)}$	$40.4^{(\uparrow 1.3)}$	$11.6^{(\uparrow 0.3)}$	$8.8^{(\uparrow 1.5)}$	$19.4^{(\uparrow 0.9)}$	$20.4^{(\uparrow 0.6)}$	$22.1^{(\uparrow 0.3)}$

TABLE IV

RESULTS ON AI-TOD V2. NOTE THAT MODELS ARE TRAINED ON THE "TRAINVAL" SET AND VALIDATED ON THE "TEST" SET. THE BEST SCORES ARE HIGHLIGHTED IN BOLD. GRAY ROWS REPRESENT THE RESULTS WITH SWIN

Method	Backbone	AP	$AP_{0.5}$	$AP_{0.75}$	AP_{vt}	AP_t	AP_s	AP_m
TridentNet [32]	ResNet-50	10.1	24.5	6.7	0.1	6.3	19.8	31.9
DotD [48]	ResNet-50	20.4	51.4	12.3	8.5	21.1	24.6	30.4
Faster R-CNN [65]	ResNet-50	12.8	29.9	9.4	0.0	9.2	24.6	37.0
Faster R-CNN w/ SWIN	ResNet-50	$14.3^{(\uparrow 1.5)}$	$31.7^{(\uparrow 1.8)}$	$11.5^{(\uparrow 2.1)}$	$0.4^{(\uparrow 0.4)}$	$10.9^{(\uparrow 1.7)}$	$25.4^{(\uparrow 0.8)}$	$38.1^{(\uparrow 1.1)}$
Faster R-CNN w/NWD-RKA [49]	ResNet-50	21.4	53.2	12.5	7.7	20.7	26.8	35.2
Faster R-CNN w/RFLA [23]	ResNet-50	22.6	54.8	14.5	8.6	21.7	28.8	35.5
Faster R-CNN w/RFLA w/ SWIN	ResNet-50	$23.5^{(\uparrow 0.9)}$	$57.2^{(\uparrow 2.4)}$	$15.6^{(\uparrow 1.1)}$	$8.7^{(\uparrow 0.1)}$	$21.5^{(\downarrow 0.2)}$	$30.2^{(\uparrow 1.4)}$	$37.0^{(\uparrow 1.5)}$
Cascade -RCNN [69]	ResNet-50	15.1	34.2	11.2	0.1	11.5	26.7	38.5
Cascade R-CNN w/ SWIN	ResNet-50	$16.5^{(\uparrow 1.4)}$	$36.7^{(\uparrow 2.5)}$	$11.4^{(\uparrow 0.2)}$	$2.4^{(\uparrow 2.3)}$	$14.6^{(\uparrow 3.1)}$	$26.9^{(\uparrow 0.2)}$	$39.7^{(\uparrow 1.2)}$
Cascade -RCNN w/NWD-RKA [49]	ResNet-50	22.2	52.5	15.1	7.8	21.8	28.0	37.2
Cascade R-CNN w/NWD-RKA w/ SWIN	ResNet-50	$23.3^{(\uparrow 1.1)}$	$55.6^{(\uparrow 3.1)}$	$15.9^{(\uparrow 0.8)}$	$8.0^{(\uparrow 1.2)}$	$23.6^{(\uparrow 1.8)}$	$28.4^{(\uparrow 0.4)}$	$37.3^{(\uparrow 0.1)}$
DetectoRS [12]	ResNet-50	16.1	35.5	12.5	0.1	12.6	28.3	40.0
DetectoRS w/ SWIN	ResNet-50	$16.8^{(\uparrow 0.7)}$	$38.6^{(\uparrow 3.1)}$	$13.7^{(\uparrow 1.2)}$	$3.3^{(\uparrow 3.2)}$	$13.1^{(\uparrow 0.5)}$	$29.4^{(\uparrow 1.1)}$	$40.4^{(\uparrow 0.4)}$
DetectoRS w/NWD-RKA [49]	ResNet-50	24.7	57.4	17.1	9.7	24.2	29.8	39.3
DetectoRS w/RFLA [23]	ResNet-50	25.7	58.9	18.8	9.2	25.5	30.2	40.2
DetectoRS w/RFLA w/ SWIN	ResNet-50	$\mathbf{26.5}^{(\uparrow 0.8)}$	$\mathbf{61.5}^{(\uparrow 2.6)}$	$\mathbf{18.8}^{(\uparrow 0)}$	$\mathbf{11.5}^{(\uparrow 2.3)}$	$\mathbf{26.2}^{(\uparrow 0.7)}$	$\mathbf{32.0}^{(\uparrow 1.8)}$	$\mathbf{40.9}^{(\uparrow 0.7)}$
YOLOv3 [77]	DarkNet-53	4.1	14.6	0.9	1.1	4.8	7.7	8.0
Retinanet [58]	ResNet-50	8.9	24.2	4.6	2.7	8.4	13.1	20.4
Retinanet w/ SWIN	ResNet-50	$9.9^{(\uparrow 1.0)}$	$26.3^{(\uparrow 2.1)}$	$5.5^{(\uparrow 0.9)}$	$4.6^{(\uparrow 1.9)}$	$8.6^{(\uparrow 0.2)}$	$13.5^{(\uparrow 0.4)}$	$20.9^{(\uparrow 0.5)}$
Retinanet w/NWD-RKA [49]	ResNet-50	10.5	28.5	5.2	3.5	11.0	15.2	24.2
Retinanet w/NWD-RKA w/ SWIN	ResNet-50	$12.6^{(\uparrow 2.1)}$	$32.0^{(\uparrow 3.5)}$	$6.3^{(\uparrow 1.1)}$	$4.9^{(\uparrow 1.4)}$	$10.2^{(\downarrow 0.8)}$	$16.9^{(\uparrow 1.7)}$	$25.0^{(\uparrow 0.8)}$
RepPoints [73]	ResNet-50	9.3	23.6	5.4	2.8	10.0	12.3	18.9
FoveaBox [78]	ResNet-50	11.3	28.1	7.4	1.4	8.6	17.8	32.2
Grid R-CNN [79]	ResNet-50	14.3	31.1	11.0	0.1	11.0	25.7	36.7
FCOS [76]	ResNet-50	12.0	30.2	7.3	2.2	11.1	16.6	26.9
FCOS w/ SWIN	ResNet-50	$13.5^{(\uparrow 1.5)}$	$32.1^{(\uparrow 1.9)}$	$8.1^{(\uparrow 0.8)}$	$5.1^{(\uparrow 2.9)}$	$14.4^{(\uparrow 4.3)}$	$16.9^{(\uparrow 0.3)}$	$27.7^{(\uparrow 0.8)}$

SOTA competitors by 0.8 and 2.6 points, respectively. SWIN consistently brings AP improvements ranging from 0.7 to 2.1 across other models. Comparing Tables III and IV, we have two observations.

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TABLE V

RESULTS ON DOTA V2. NOTE THAT MODELS ARE TRAINED ON THE "TRAIN" SET AND VALIDATED ON THE "TEST" SET. THE BEST SCORES ARE HIGHLIGHTED IN BOLD

Method	AP	$AP_{0.5}$	AP_{vt}	AP_t
FCOS	31.8	55.4	0.3	4.0
FCOS w/RFLA	32.1	55.6	0.7	6.8
Faster R-CNN	35.6	59.5	0.0	7.1
Faster R-CNN w/RFLA	36.3	61.5	1.9	11.7
DetectoRS	40.8	62.6	0.0	7.0
DetectoRS w/RFLA	41.3	64.2	2.1	10.8
FCOS w/RFLA w/ SWIN	$33.5^{(\uparrow 1.4)}$	$56.1^{(\uparrow 0.5)}$	$1.8^{(\uparrow 1.1)}$	$7.1^{(\uparrow 0.3)}$
Faster R-CNN w/RFLA w/ SWIN	$37.5^{(\uparrow 1.2)}$	$63.1^{(\uparrow 1.6)}$	$3.2^{(\uparrow 1.3)}$	$11.4^{(\downarrow 0.3)}$
DetectoRS w/RFLA w/ SWIN	$\mathbf{41.9}^{(\uparrow 0.6)}$	$\mathbf{66.3}^{(\uparrow 2.1)}$	$\mathbf{4.4}^{(\uparrow 2.3)}$	$\mathbf{12.2}^{(\uparrow 1.4)}$

TABLE VI

RESULTS OF YOLO-BASED METHODS ON AI-TOD V1. NOTE THAT MODELS ARE TRAINED ON THE "TRAINVAL" SET AND VALIDATED ON THE "TEST" SET. THE BEST SCORES ARE HIGHLIGHTED IN BOLD

Method	Based YOLO	AP	$AP_{0.5}$	AP_{vt}	AP_t
YOLOv5m	YOLOv5m	27.1	61.4	11.3	22.9
FFCA-YOLO [47]	YOLOv5m	27.7	61.7	11.7	23.2
YOLOv5m w/SWIN	YOLOv5m	**27.9**	**63.0**	**12.1**	**24.4**
MS-YOLO-s [80]	YOLOv8s	26.2	55.1	8.5	24.3
RS-YOLOV8s [81]	YOLOv8s	-	56.3	-	-
YOLOv9m [82]	YOLOv9m	26.7	61.1	10.9	24.0
YOLOv9m w/SWIN	YOLOv9m	27.4	62.8	11.5	25.1

1) The results in AI-TOD v2 are overall superior to those in AI-TOD v1. This is because AI-TOD v2 has a higher annotation quality, which leads to a fairer evaluation of the network. Therefore, all subsequent ablation experiments are conducted on the AI-TOD v2 by default.
2) The integration of SWIN leads to an overall performance improvement, especially on very tiny and tiny objects.

To prove the generalized ability of SWIN, we conduct experiments on the DOTA v2 dataset. As shown in Table V, the introduction of the SWIN results in performance improvements of 1.4, 1.2, and 0.6 points @AP for models FCOS w/RFLA, Faster R-CNN w/RFLA, and DetectoRS w/RFLA, respectively, as well as enhancements of 0.5, 1.6, and 2.1 points @$AP_{0.5}$. In terms of detecting very tiny and tiny objects, SWIN generally enhances the detection performance across networks, with improvements ranging from 0.3 to 2.3 points. The only exception is a slight decrease of 0.3 points in AP_t metric for the Faster R-CNN w/RFLA w/SWIN network. This result demonstrates the effectiveness and generalization of SWIN.

Given the outstanding performance of YOLO series methods, we embed SWIN into YOLOv5m and YOLOv9m, and compare them with several existing YOLO-based methods, as shown in Table VI. Notably, the YOLOv9m w/SWIN achieves better performance to the current best method (FFCA-YOLO). More significantly, the YOLOv5m w/SWIN achieves the highest performance, surpassing the existing best method by 1.3 points in $AP_{0.5}$, further validating the effectiveness and superiority of the proposed method in this article.

2) Ablation Study: We conduct ablation experiments on the DetectoRS w/RFLA network to evaluate the effectiveness of the proposed modules and compare them with other methods.

a) Effectiveness of individual component: As shown in Table VII, the use of the INEN module leads to performance improvements of 0.6@AP and 1.5@$AP_{0.5}$. Similarly, employing the SWL module results in improvements of 0.4@AP and 0.6@$AP_{0.5}$. Remarkably, simultaneous deployment of both the INEN module and the SWL module achieves the SOTA performance, yielding improvements of 0.8@AP and 2.6@$AP_{0.5}$ compared to the baseline. When the network uses the INEN module alone, the AP_{vt} metric achieves the best performance. More generally, when both INEN and SWL are used simultaneously, the AP_t, AP_s, and AP_m metrics achieve the best performance. This underscores the synergistic effects of these modules and their significant contributions to overall detection accuracy.

TABLE VII

ABLATION RESULTS OF INEN AND SWL MODULE. THE BEST SCORES ARE HIGHLIGHTED IN BOLD

INEN	SWL	AP	$AP_{0.5}$	$AP_{0.75}$	AP_{vt}	AP_t	AP_s	AP_m
×	×	25.7	58.9	18.8	9.2	25.5	30.2	40.2
√	×	26.3	60.4	18.8	**12.7**	26.1	31.1	40.8
×	√	26.1	59.5	**19.1**	10.5	25.7	31.0	**40.9**
√	√	**26.5**	**61.5**	18.8	11.5	**26.2**	**32.0**	**40.9**

TABLE VIII

ABLATION RESULTS OF INEN MODULE. BASELINE IS THE DETECTORS W/RFLA. THE BEST SCORES ARE HIGHLIGHTED IN BOLD

Method	AP	$AP_{0.5}$	$AP_{0.75}$	AP_{vt}	AP_t	AP_s	AP_m
Baseline	25.7	58.9	**18.8**	9.2	25.5	30.2	40.2
+ w/GCN [54]	25.8	59.2	17.5	8.9	25.8	30.5	41.0
+ w/GAN [83]	25.7	59.3	17.6	8.5	25.2	30.2	41.2
+ w/GANv2 [84]	25.9	59.4	17.8	9.1	25.7	30.4	**41.4**
+ w/INEN	**26.3**	**60.4**	**18.8**	**12.7**	**26.1**	**31.1**	40.8

b) INEN versus other GCNs: INEN constructs an asymmetric adjacency matrix based on the confidence of different proposals, thereby guiding asymmetric information propagation among instances. In contrast, we compared it with GCN [54], which uses a symmetric adjacency matrix where the weights between nodes of the same category are 1 and the weights between nodes of different categories are 0. Additionally, we evaluated the performance of GCNs with attention mechanisms, such as graph attention network (GAN) [83] and GANv2 [84]. Furthermore, we explore the integration of GCNs (including GCN, GAN, and GANv2) and INEN into the Head structure.

From Table VIII, both GCNs and INEN improve the performance of the network. Specifically, GCN slightly improves the AP and $AP_{0.5}$ by 0.1 and 0.3 points, while GANv2 shows improvements of 0.2 and 0.5 points, respectively. In contrast, INEN achieves significant improvements of 0.6 and 1.5 points for AP and $AP_{0.5}$. We attribute these improvements to the information exchange capability of these modules. In comparison, the performance enhancement from the INEN model is more significant, which we believe is due to its ability to filter and exchange more effective feature information. Conversely, although the attention mechanisms in GAN and GANv2 can guide the aggregation of effective information, this aggregation

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TABLE IX

COMPARISON OF DISTANCE METRICS. NOTE THAT "sL1" MEANS SMOOTHL1LOSS. THE BEST SCORES ARE HIGHLIGHTED IN BOLD

Loss	AP	$AP_{0.5}$	$AP_{0.75}$	AP_{vt}	AP_t	AP_s	AP_m
sL1	25.7	58.9	18.8	9.2	25.5	30.2	40.2
IOUloss	23.9	55.9	16.5	8.6	23.1	29.7	40.5
NWDloss	24.8	57.8	17.4	9.1	25.4	28.7	39.9
SWL	**26.1**	**59.5**	**19.1**	**10.5**	**25.7**	**31.0**	**40.9**

TABLE X

ABLATION RESULTS OF SWL MODULE. NOTE THAT "sL1" MEANS SMOOTHL1LOSS. THE BEST SCORES ARE HIGHLIGHTED IN BOLD

RPN	Head	AP	$AP_{0.5}$	$AP_{0.75}$	AP_{vt}	AP_t	AP_s	AP_m
sL1	sL1	25.7	58.9	18.8	9.2	25.5	30.2	40.2
SWL	sL1	**26.1**	59.3	19.0	10.3	25.6	30.5	**41.0**
sL1	SWL	26.0	58.8	**19.2**	**10.6**	25.1	29.8	40.7
SWL	SWL	**26.1**	**59.5**	19.1	10.5	**25.7**	**31.0**	40.9

capability proves to be suboptimal for tiny targets with very few features.

c) SWL versus other metrics: We conduct comparative experiments by evaluating norm-based smoothL1loss, IoU-based IoUloss, distribution-based NWDloss, and the SWL proposed in this study. As shown in Table IX, model with SWL achieves the best performance with 26.1@AP and 59.5@$AP_{0.5}$, respectively. Notably, the model employing SWL demonstrates a 1.3 points increase in AP_{vt}, underscoring the effectiveness of the SWL metric for TOD.

To validate the effectiveness of SWL in a neural network, we replaced the default smoothL1loss with SWL at different positions (the RPN and Head structures) of the DetectoRS network. The experimental results, shown in Table X, demonstrate the performance of the SWL module in various position regression structures. The experimental results demonstrate that achieving the best performance is achieved when SWL is applied simultaneously to both the RPN and Head structures.

3) Discussion: We conduct comparative experiments on hyperparameters, including the values of the blending coefficients α and the threshold Thres. Furthermore, to offer a comprehensive illustration of the performance of our method, we conduct a statistical analysis of computation speed and parameter size on several networks before and after the integration of our proposed module.

a) Performance of different α: As in (1), α is utilized to balance the intensity of loss from two classification branches. In Table XI, we conduct experiments on DetectoRS w/RFLA w/SWIN, keeping all other parameters fixed and experimentally show that the network achieves the best performance when α is set to 0.7.

b) Performance of different Thres: As in (14), Thres dictates the threshold for the piecewise computation of the loss function. Similar to α, we conduct experiments on DetectoRS w/RFLA w/SWIN, keeping all other parameters fixed. In Table XII, the network achieves the best performance when Thres is set to 1/9 for RPN and 1.0 for Head, respectively.

c) Computational analysis: As shown in Table XIII, integrating SWIN into two-stage networks such as Faster R-CNN w/RFLA, Cascade R-CNN w/RFLA, and DetectoRS w/RFLA results in only a marginal increase in the number of parameters, from 41.16 to 41.23M, 68.95 to 69.19M, and 123.22 to 125.26M, respectively. This is accompanied by an increase in "FLOPs" ranging from 1.02 to 3.06 GFLOPs. Concurrently, there is a slight reduction in FPS, from 18.79 to 18.53, 14.88 to 13.96, and 8.54 to 8.34, respectively. Similarly, incorporating SWIN into one-stage networks such as YOLOv5 and YOLOv9 results in an increased computational cost. Specifically, FLOPs for YOLOv5 increase from 75.48 to 78.8G, while FLOPs for YOLOv9 rise from 133 to 177.32G. This increase in computational load impacts computational speed, with FPS decreasing from 29.26 to 24.63 for YOLOv5 and from 19.63 to 12.85 for YOLOv9. The increase in FLOPs and the decrease in FPS are attributed to the dual computational burden of the Head component in YOLO networks and the additional computational load introduced by the INEN module. This increased computational burden highlights the necessity to optimize the integration of SWIN.

TABLE XI

INFLUENCE OF THE BLENDING COEFFICIENTS α. NOTE THAT THE EXPERIMENT IS CONDUCTED ON DETECTORS W/RFLA W/SWIN. THE BEST SCORES ARE HIGHLIGHTED IN BOLD

α	AP	$AP_{0.5}$	AP_{vt}	AP_t
0.65	26.4	60.7	**12.6**	25.7
0.7	**26.5**	**61.5**	11.5	**26.2**
0.75	26.3	60.3	12.4	25.7
0.8	26.2	59.9	12.5	25.5

TABLE XII

INFLUENCE OF THE THRES. NOTE THAT THE EXPERIMENT IS CONDUCTED ON DETECTORS W/RFLA W/SWIN. THE BEST SCORES ARE HIGHLIGHTED IN BOLD

RPN	Head	AP	$AP_{0.5}$	AP_{vt}	AP_t
1/3	0.5	25.3	59.4	9.7	25.1
1/3	1.0	25.6	60.6	9.9	25.4
1/3	1.5	25.4	59.8	10.0	25.9
1/9	0.5	26.4	60.7	**11.6**	25.8
1/9	1.0	**26.5**	**61.5**	11.5	**26.2**
1/9	1.5	26.3	61.4	10.9	26.1
1/18	0.5	25.7	**61.5**	10.1	25.4
1/18	1.0	26.0	59.5	10.5	26.1
1/18	1.5	26.1	60.4	9.7	25.3

TABLE XIII

STATISTICAL COMPARISON OF FPS, FLOPS, AND PARAMETERS

Method	FPS	FLOPs	Parameters
Faster R-CNN w/RFLA	18.79	134.42G	41.16M
Faster R-CNN w/RFLA w/SWIN	18.53	135.44G	41.23M
Cascade R-CNN w/RFLA	14.88	162.21G	68.95M
Cascade R-CNN w/RFLA w/SWIN	13.96	165.27G	69.19M
DetectoRS w/RFLA	8.54	159.68G	123.22M
DetectoRS w/RFLA w/SWIN	8.34	162.74G	125.26M
YOLOv5m	29.26	75.48G	19.93M
YOLOv5m w/SWIN	24.63	78.80G	20.17M
YOLOv9m	19.63	133.0G	31.36M
YOLOv9m w/SWIN	12.85	177.32G	31.50M

C. Visualization and Analysis

1) Visualization of High-Quality Results: We visualize the inference results of our proposed method applied to Faster

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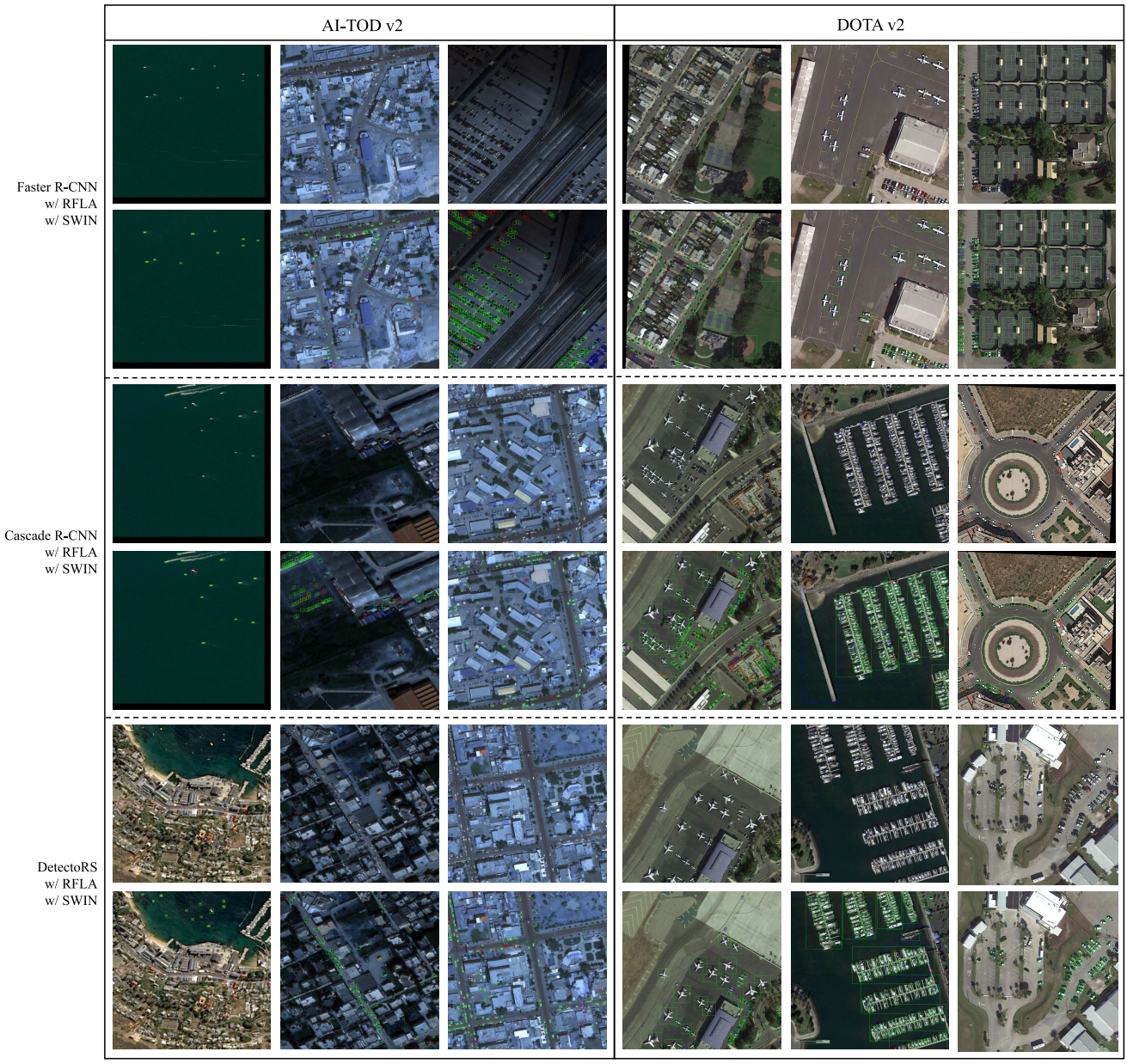


Fig. 7. High-quality results on AI-TOD v2 and DOTA v2. Green boxes represent TP, red boxes represent FN, and blue boxes represent FP. The first, third, and fifth lines represent the original images, whereas the second, fourth, and sixth lines depict the inference results of networks Faster R-CNN w/RFLA w/SWIN, Cascade R-CNN w/RFLA w/SWIN, and DetectoRS w/RFLA w/SWIN, respectively.

R-CNN w/RFLA, Cascade R-CNN w/RFLA, and DetectoRS w/RFLA baseline models (with IoU threshold of 0.5 and class confidence threshold of 0.5), as depicted in Figs. 7–10. Fig. 7 illustrates the typical recognition results of the proposed method on AI-TOD v2 and DOTA v2; the first, third, and fifth lines represent the original images, whereas the second, fourth, and sixth lines depict the inference results of networks Faster R-CNN w/RFLA w/SWIN, Cascade R-CNN w/RFLA w/SWIN, and DetectoRS w/RFLA w/SWIN, respectively. In the recognition results, green represents TP, red represents false negative (FN), and blue represents false positive (FP). It can be observed from the recognition results that the proposed method achieves high-quality recognition performance on these datasets. Furthermore, to visually demonstrate the effectiveness of the proposed approach, inference results are compared in Figs. 8–10. In Figs. 8–10, each figure presents three rows: the first row shows the GT labels, the second row shows the inference results of the baseline, and the third row displays the results of the models modified by our proposed method. From Figs. 8–10, it is evident that our method effectively reduces FN predictions, which is the main reason of the improvement.

Fig. 11 visualizes the inference results alongside activation heatmaps of DetectoRS w/ RFLA baseline and the modified DetectoRS w/ RFLA with our method. The first column represents GT, while the second and third columns depict the inference results and activation heatmaps of DetectoRS w/ RFLA. The fourth and fifth columns show the inference results and activation heatmaps of DetectoRS w/ RFLA w/SWIN. Comparing the activation heatmaps reveals that our method induces the network to focus more attention on GT regions with salient features. More specifically, in the images of the

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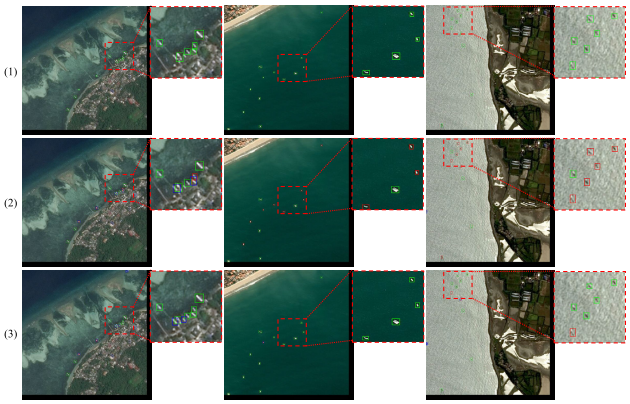


Fig. 8. Visualization results on AI-TOD v2. The first row is the GT, the second row is the result of Faster R-CNN w/ RFLA, and the third row is the result of the modified Faster R-CNN w/ RFLA with SWIN. Green boxes represent TP, red boxes represent FN, and blue boxes represent FP.



Fig. 9. Visualization results on AI-TOD v2. The first row is the GT, the second row is the result of Cascade R-CNN w/ RFLA, and the third row is the result of the modified Cascade R-CNN w/ RFLA with SWIN. Green boxes represent TP, red boxes represent FN, and blue boxes represent FP.

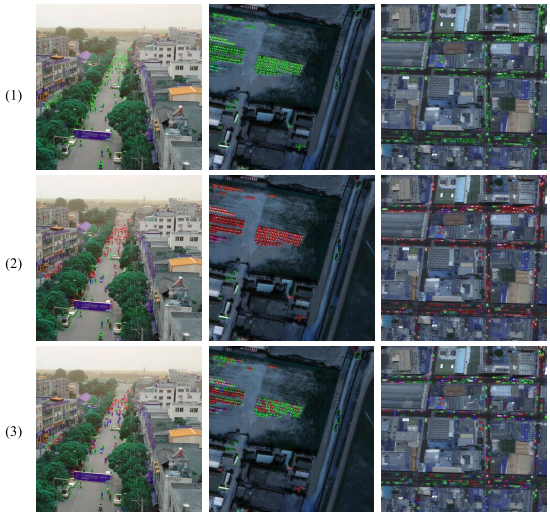


Fig. 10. Visualization results on AI-TOD v2. The first row is the GT, the second row is the result of DetectoRS w/ RFLA, and the third row is the result of the modified DetectoRS w/ RFLA with SWIN. Green boxes represent TP, red boxes represent FN, and blue boxes represent FP.

first row, the introduction of the SWIN model results in enhanced activation values for "*ship* (SH)" category instances, while simultaneously suppressing activation values associated with background regions such as the coast. This ability of the

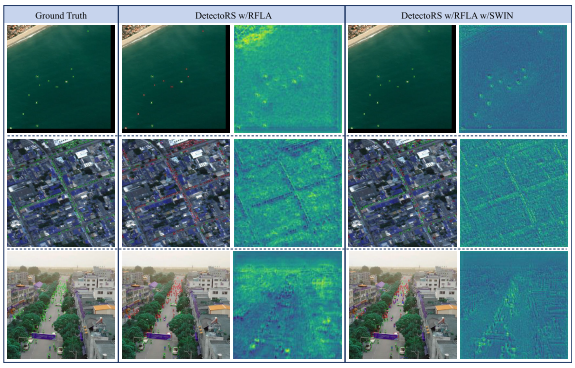


Fig. 11. Visualization of inference results and activation heatmaps. The first column represents GT, while the second and third columns depict the inference results and activation heatmaps of DetectoRS w/ RFLA. The fourth and fifth columns show the inference results and activation heatmaps of DetectoRS w/ RFLA w/SWIN. The activation heatmaps reveal that SWIN induces the network to focus more attention on GT regions with salient features.

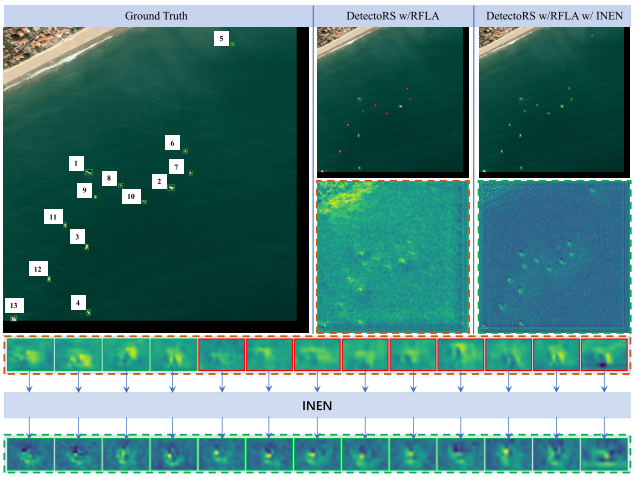


Fig. 12. Visualization of INEN effectiveness. At the top, from left to right, the respective components are GT, inference results and activation heatmaps of DetectoRS w/RFLA, and inference results and activation heatmaps of DetectoRS w/RFLA w/SWIN. At the bottom, the activation heatmap of the GT region on DetectoRS w/RFLA and DetectoRS w/RFLA w/SWIN is enlarged.

SWIN model is further demonstrated in the subsequent rows, where similar effects are observed for instances of the "*vehicle* (VE)" and "*person* (PE)" categories, respectively.

2) Visualization of INEN Improvement: Incorporating the INEN module into the DetectoRS w/RFLA baseline model, as depicted in Fig. 12, visualizes the inference results alongside activation heatmaps. At the top of Fig. 12, from left to right, the respective components are GT, inference results and activation heatmaps of DetectoRS w/RFLA, and inference results and activation heatmaps of DetectoRS w/RFLA w/SWIN. At the bottom of Fig. 12, regions outlined with red dashed boxes indicate the enlarged activation heatmap of the GT region on DetectoRS w/RFLA, solid red boxes denote FN, while green dashed boxes highlight the enlarged activation heatmap of the GT region on DetectoRS w/RFLA w/SWIN. Comparing the heatmaps before and after adding the INEN module reveals that the network with INEN integration enhances weakly activated proposals to levels comparable to strongly activated ones. This enhancement significantly reduces FN instances.

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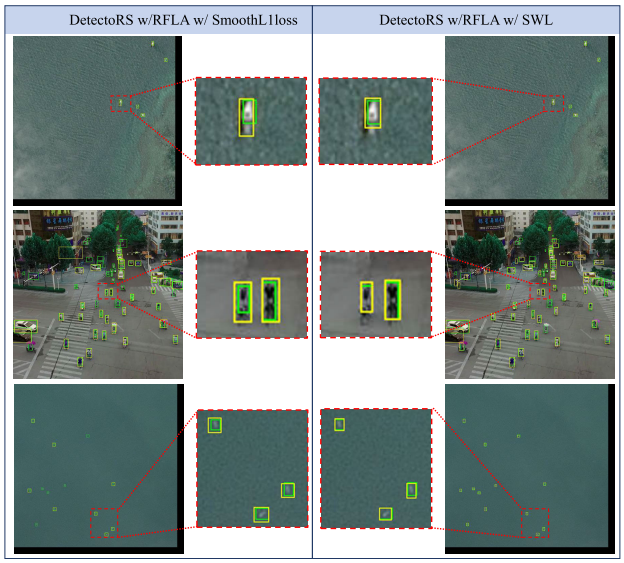


Fig. 13. Visualization of boxes regression with different regression loss. The yellow boxes represent the predicted bounding boxes, while the green boxes represent the GT.

3) Visualization of SWL Effectiveness: Fig. 13 illustrates the visualization of the inference results obtained by DetectoRS w/RFLA using smoothL1loss and SWL. Specifically, we have zoomed in on the position regression section. The left side depicts the position regression results using smoothL1loss, while the right side shows the position regression results using SWL. It can be observed from Fig. 13 that the network using SWL achieves a more precise localization accuracy.

D. Limitations and Future Work

The limitations of our study include the non-scale-invariance of Wasserstein distance, potentially biasing the network toward larger targets due to higher associated losses. In contrast, scale-invariant metrics such as KL divergence provide partial relief, but struggle with accurately representing distances when distributions significantly differ (low overlap). Addressing these challenges to achieve both scale invariance and effective distance representation represents a critical future research direction for our work. The method proposed in this article not only demonstrates potential for performance enhancement across multiple baseline models but also introduces a computational burden. Future research will focus on developing lightweight integration strategies within various frameworks to alleviate this computational cost.

Meanwhile, enlarging images not only appropriately significantly enhances performance but also presents an exponential increase in computational costs. Balancing performance and cost-effectiveness is crucial for future developments. Regarding TOD, COCO's evaluation metrics (IoU and AP) may not fully assess this task due to their strict requirements for extremely small instances. Developing customized evaluation metrics that prioritize object recognition and approximate localization over strict accuracy is particularly important. Drawing insights from fields such as crowd counting, to devise suitable evaluation metrics for guiding object detection architectures in specific scenarios, will play a crucial role in future research development. Since transformers have demonstrated outstanding performance in visual tasks, employing their self-attention mechanisms as an alternative to the INEN approach proposed in this article for feature fusion between nodes represents a promising avenue for exploration.

V. Conclusion

Starting from the recognition results of classical networks, this article identifies existing problems in TOD. Analyzing the features of tiny objects reveals two shortcomings of traditional methods in TOD tasks: inadequate extraction of discriminative features and inappropriate regression measurement metrics. Subsequently, addressing these issues, we propose the INEN and the SWL distance metric. Furthermore, efforts are made to theoretically explain the validity of these methods. Extensive comparative experiments and ablation studies are conducted on the popular TOD datasets AI-TOD v1/v2 and DOTA v2. The results demonstrate the effectiveness of the proposed methods. Additionally, visualization analysis further demonstrates the effectiveness of this work.

References

[1] G.-S. Xia et al., "DOTA: A large-scale dataset for object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2018, pp. 3974–3983.

[2] Z. Zou, K. Chen, Z. Shi, Y. Guo, and J. Ye, "Object detection in 20 years: A survey," *Proc. IEEE*, vol. 111, no. 3, pp. 257–276, Mar. 2023.

[3] J. Wang, J. Ding, H. Guo, W. Cheng, T. Pan, and W. Yang, "Mask OBB: A semantic attention-based mask oriented bounding box representation for multi-category object detection in aerial images," *Remote Sens.*, vol. 11, no. 24, p. 2930, Dec. 2019.

[4] Y. Lin, H. Sun, N. Liu, Y. Bian, J. Cen, and H. Zhou, "A lightweight multi-scale context network for salient object detection in optical remote sensing images," in *Proc. 26th Int. Conf. Pattern Recognit. (ICPR)*, Aug. 2022, pp. 238–244.

[5] J. Wang, W. Yang, H. Guo, R. Zhang, and G.-S. Xia, "Tiny object detection in aerial images," in *Proc. 25th Int. Conf. Pattern Recognit. (ICPR)*, Jan. 2021, pp. 3791–3798.

[6] T. Lin et al., "Microsoft COCO: Common objects in context," in *Proc. Eur. Conf. Comput. Vis.*, 2014, pp. 740–755.

[7] Y. Wan, Z. Liao, J. Liu, W. Song, H. Ji, and Z. Gao, "Small object detection leveraging density-aware scale adaptation," *Photogrammetric Rec.*, vol. 38, no. 182, pp. 160–175, Jun. 2023.

[8] J. Chen, K. Chen, H. Chen, Z. Zou, and Z. Shi, "A degraded reconstruction enhancement-based method for tiny ship detection in remote sensing images with a new large-scale dataset," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5625014.

[9] Z. Zou and Z. Shi, "Random access memories: A new paradigm for target detection in high resolution aerial remote sensing images," *IEEE Trans. Image Process.*, vol. 27, no. 3, pp. 1100–1111, Mar. 2018.

[10] X. Yang et al., "Optical remote sensing ship small target detection based on up-CBAM-RYOLO v5," *J. Jilin Univ. Inf. Sci. Ed.*, early access, pp. 1–10, Mar. 2024, doi: 10.19292/j.cnki.jdxxp.20240305.001.

[11] Z. Zeng et al., "RCSANet: A full convolutional network for extracting inland aquaculture ponds from high-spatial-resolution images," *Remote Sens.*, vol. 13, no. 1, p. 92, Dec. 2020.

[12] S. Qiao, L.-C. Chen, and A. Yuille, "DetectoRS: Detecting objects with recursive feature pyramid and switchable atrous convolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 10213–10224.

[13] Y. Xiang et al., "Progress in small object detection for remote sensing images," *J. Image Graph.*, vol. 2023, no. 6, pp. 1662–1684, 2023.

[14] Q. Liu, R. Liu, B. Zheng, H. Wang, and Y. Fu, "Infrared small target detection with scale and location sensitivity," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 17490–17499.

[15] T. Wu et al., "MTU-Net: Multilevel TransUnet for space-based infrared tiny ship detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5601015.

[16] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 779–788.

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[17] Q. Li, Z. Han, and X. Wu, "Deeper insights into graph convolutional networks for semi-supervised learning," in *Proc. AAAI Conf. Artif. Intell.*, 2018, pp. 3538–3545.

[18] Z. Wu, S. Pan, F. Chen, G. Long, C. Zhang, and P. S. Yu, "A comprehensive survey on graph neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 1, pp. 4–24, Jan. 2021.

[19] R. Girshick, "Fast R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1440–1448.

[20] J. Wang, C. Xu, W. Yang, and L. Yu, "A normalized Gaussian Wasserstein distance for tiny object detection," 2022, *arXiv:2110.13389*.

[21] Z. Liu, Z. Jiang, W. Feng, and H. Feng, "OD-GCN: Object detection boosted by knowledge GCN," in *Proc. IEEE Int. Conf. Multimedia Expo Workshops (ICMEW)*, Jul. 2020, pp. 1–6.

[22] Y. Ma, L. Chai, L. Jin, and J. Yan, "Hierarchical alignment network for domain adaptive object detection in aerial images," *ISPRS J. Photogramm. Remote Sens.*, vol. 208, pp. 39–52, Feb. 2024.

[23] C. Xu, J. Wang, W. Yang, H. Yu, L. Yu, and G.-S. Xia, "RFLA: Gaussian receptive field based label assignment for tiny object detection," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 526–543.

[24] M. Kisantal, Z. Wojna, J. Murawski, J. Naruniec, and K. Cho, "Augmentation for small object detection," in *Proc. Int. Conf. Adv. Comput. Inf. Technol.*, 2019, pp. 1–15.

[25] B. Bosquet, D. Cores, L. Seidenari, V. M. Brea, M. Mucientes, and A. D. Bimbo, "A full data augmentation pipeline for small object detection based on generative adversarial networks," *Pattern Recognit.*, vol. 133, Jan. 2023, Art. no. 108998.

[26] Y. Bai, Y. Zhang, M. Ding, and B. Ghanem, "SOD-MTGAN: Small object detection via multi-task generative adversarial network," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Jun. 2018, pp. 210–226.

[27] J. Li, X. Liang, Y. Wei, T. Xu, J. Feng, and S. Yan, "Perceptual generative adversarial networks for small object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 1951–1959.

[28] J. Noh, W. Bae, W. Lee, J. Seo, and G. Kim, "Better to follow, follow to be better: Towards precise supervision of feature super-resolution for small object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9724–9733.

[29] J. Zhang, J. Lei, W. Xie, Z. Fang, Y. Li, and Q. Du, "SuperYOLO: Super resolution assisted object detection in multimodal remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5605415.

[30] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 936–944.

[31] S. Liu, L. Qi, H. Qin, J. Shi, and J. Jia, "Path aggregation network for instance segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8759–8768.

[32] Y. Li, Y. Chen, N. Wang, and Z.-X. Zhang, "Scale-aware trident networks for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6053–6062.

[33] K. Sun, B. Xiao, D. Liu, and J. Wang, "Deep high-resolution representation learning for human pose estimation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Oct. 2019, pp. 5693–5703.

[34] M. Tan, R. Pang, and Q. V. Le, "EfficientDet: Scalable and efficient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 10778–10787.

[35] H. Hu, J. Gu, Z. Zhang, J. Dai, and Y. Wei, "Relation networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2018, pp. 3588–3597.

[36] X. Tang, D. K. Du, Z. He, and J. Liu, "PyramidBox: A context-assisted single shot face detector," in *Proc. ECCV*, 2018, pp. 812–828.

[37] G. Zhang, S. Lu, and W. Zhang, "CAD-Net: A context-aware detection network for objects in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 10015–10024, Dec. 2019.

[38] X. Lu, J. Ji, Z. Xing, and Q. Miao, "Attention and feature fusion SSD for remote sensing object detection," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–9, 2021.

[39] Y. Li, Q. Huang, X. Pei, Y. Chen, L. Jiao, and R. Shang, "Cross-layer attention network for small object detection in remote sensing imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 2148–2161, 2021.

[40] B. Singh and L. S. Davis, "An analysis of scale invariance in object detection—SNIP," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3578–3587.

[41] B. Singh, M. Najibi, and L. Davis, "SNIPER: Efficient multi-scale training," in *Proc. Neural Inf. Process. Syst. (NeurIPS)*, 2018, pp. 9333–9343.

[42] M. Najibi, B. Singh, and L. Davis, "AutoFocus: Efficient multi-scale inference," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9744–9754.

[43] Y. Kim, B.-N. Kang, and D. Kim, "SAN: Learning relationship between convolutional features for multi-scale object detection," in *Proc. ECCV*, 2018, pp. 328–343.

[44] Y. Jiang et al., "R2CNN: Rotational region CNN for orientation robust scene text detection," 2017, *arXiv:1706.09579*.

[45] J. Wu, Z. Pan, B. Lei, and Y. Hu, "FSANet: Feature-and-spatial-aligned network for tiny object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5630717.

[46] S. Li, Q. Tong, X. Liu, Z. Cui, and X. Liu, "MA2-FPN for tiny object detection from remote sensing images," in *Proc. 15th Int. Congr. Image Signal Process., Biomed. Eng. Informat. (CISP-BMEI)*, Nov. 2022, pp. 1–8.

[47] Y. Zhang, M. Ye, G. Zhu, Y. Liu, P. Guo, and J. Yan, "FFCA-YOLO for small object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5611215.

[48] C. Xu, J. Wang, W. Yang, and L. Yu, "Dot distance for tiny object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2021, pp. 1192–1201.

[49] C. Xu, J. Wang, W. Yang, H. Yu, L. Yu, and G.-S. Xia, "Detecting tiny objects in aerial images: A normalized Wasserstein distance and a new benchmark," *ISPRS J. Photogramm. Remote Sens.*, vol. 190, pp. 79–93, Aug. 2022.

[50] F. Scarselli, M. Gori, A. C. Tsoi, M. Hagenbuchner, and G. Monfardini, "The graph neural network model," *IEEE Trans. Neural Netw.*, vol. 20, no. 1, pp. 61–80, Jan. 2009.

[51] L. Wang, Y. Huang, Y. Hou, S. Zhang, and J. Shan, "Graph attention convolution for point cloud semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 10296–10305.

[52] H. Ren et al., "Graph convolutional networks in language and vision: A survey," *Knowl.-Based Syst.*, vol. 251, Sep. 2022, Art. no. 109250.

[53] P. Yin, J. Ye, G. Lin, and Q. Wu, "Graph neural network for 6D object pose estimation," *Knowl.-Based Syst.*, vol. 218, Apr. 2021, Art. no. 106839.

[54] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2017, pp. 1–14.

[55] L. Wang et al., "SAT-GCN: Self-attention graph convolutional network-based 3D object detection for autonomous driving," *Knowl.-Based Syst.*, vol. 259, Jan. 2023, Art. no. 110080.

[56] L. Wang, C. Wang, X. Zhang, T. Lan, and J. Li, "S-AT GCN: Spatial-attention graph convolution network based feature enhancement for 3D object detection," 2021, *arXiv:2103.08439*.

[57] G. Zhao, W. Ge, and Y. Yu, "GraphFPN: Graph feature pyramid network for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 2743–2752.

[58] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2999–3007.

[59] H. Zhang, Y. Wang, F. Dayoub, and N. Sunderhauf, "VarifocalNet: An IoU-aware dense object detector," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 8510–8519.

[60] J. Yu, Y. Jiang, Z. Wang, Z. Cao, and T. Huang, "UnitBox: An advanced object detection network," in *Proc. 24th ACM Int. Conf. Multimedia*, Oct. 2016, pp. 516–520.

[61] H. Rezatofighi, N. Tsoi, J. Gwak, A. Sadeghian, I. Reid, and S. Savarese, "Generalized intersection over union: A metric and a loss for bounding box regression," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 658–666.

[62] Z. Zheng, P. Wang, W. Liu, J. Li, R. Ye, and D. Ren, "Distance-IoU loss: Faster and better learning for bounding box regression," in *Proc. AAAI Conf. Artif. Intell.*, Apr. 2020, vol. 34, no. 7, pp. 12993–13000.

[63] Y.-F. Zhang, W. Ren, Z. Zhang, Z. Jia, L. Wang, and T. Tan, "Focal and efficient IOU loss for accurate bounding box regression," *Neurocomputing*, vol. 506, pp. 146–157, Sep. 2022.

[64] J. Wang, W. Yang, H.-C. Li, H. Zhang, and G.-S. Xia, "Learning center probability map for detecting objects in aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4307–4323, May 2021.

[65] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.

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[66] J. Ding et al., "Object detection in aerial images: A large-scale benchmark and challenges," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 11, pp. 7778–7796, Nov. 2022.
[67] K. Chen et al., "MMDetection: Open MMLab detection toolbox and benchmark," 2019, *arXiv:1906.07155*.
[68] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[69] Z. Cai and N. Vasconcelos, "Cascade R-CNN: Delving into high quality object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6154–6162.
[70] W. Liu et al., "SSD: Single shot MultiBox detector," in *Proc. ECCV*, 2016, pp. 21–37.
[71] K. Kim and H. S. Lee, "Probabilistic anchor assignment with IoU prediction for object detection," in *Proc. ECCV*, 2020, pp. 355–371.
[72] S. Zhang, C. Chi, Y. Yao, Z. Lei, and S. Z. Li, "Bridging the gap between anchor-based and anchor-free detection via adaptive training sample selection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9756–9765.
[73] Z. Yang, S. Liu, H. Hu, L. Wang, and S. Lin, "RepPoints: Point set representation for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9656–9665.
[74] Z. Ge, S. Liu, Z. Li, O. Yoshie, and J. Sun, "OTA: Optimal transport assignment for object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 303–312.
[75] B. Zhu et al., "AutoAssign: Differentiable label assignment for dense object detection," 2020, *arXiv:2007.03496*.
[76] Z. Tian, C. Shen, H. Chen, and T. He, "FCOS: Fully convolutional one-stage object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9626–9635.
[77] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," 2018, *arXiv:1804.02767*.
[78] T. Kong, F. Sun, H. Liu, Y. Jiang, L. Li, and J. Shi, "FoveaBox: Beyound anchor-based object detection," *IEEE Trans. Image Process.*, vol. 29, pp. 7389–7398, 2020.
[79] X. Lu, B. Li, Y. Yue, Q. Li, and J. Yan, "Grid R-CNN," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 7363–7372.
[80] X. Cao, M. Duan, H. Ding, and Z. Yang, "MS-YOLO: Integration-based multi-subnets neural network for object detection in aerial images," *Earth Sci. Informat.*, vol. 17, no. 3, pp. 2085–2106, Jun. 2024.
[81] S. Xie, M. Zhou, C. Wang, and S. Huang, "CSPPartial-YOLO: A lightweight YOLO-based method for typical objects detection in remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 388–399, 2024.
[82] C.-Y. Wang, I.-H. Yeh, and H.-Y. M. Liao, "YOLOv9: Learning what you want to learn using programmable gradient information," 2024, *arXiv:2402.13616*.
[83] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, "Graph attention networks," 2017, *arXiv:1710.10903*.
[84] S. Brody, U. Alon, and E. Yahav, "How attentive are graph attention networks?" in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2022, pp. 1–26.



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